

MIHIR BRIJESH SOLANKI

MSc Applied Cyber Security | CompTIA Security+ | Network Defense | Application Security | Ethical Hacking

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PROFESSIONAL SUMMARY

CompTIA Security+ (SY0-701) certified cybersecurity professional pursuing an MSc in Applied Cyber Security at Queen's University Belfast (Russell Group), backed by 8+ industry certifications across CompTIA and Cisco Networking Academy. Hands-on MSc projects span secure software engineering (hardened Flask REST API with full DevSecOps CI/CD pipeline and 90% test coverage), AI security (adversarial ML red/blue team assessment using FGSM, PGD, model stealing, and adversarial training), and penetration testing (RCE exploitation against a simulated Linux web server via CWE-78 OS Command Injection). Combines offensive and defensive security skill with 6 months of MERN-stack development experience implementing secure authentication, RBAC, and protected REST APIs — a developer-to-defender perspective ideal for Application Security, DevSecOps, SOC, or Penetration Testing roles. Proficient with Kali Linux, Nmap, Gobuster, Wireshark, PyTorch, and Bandit; OWASP Top 10 and CWE-aware.

CORE COMPETENCIES

Penetration Testing • Vulnerability Assessment & Exploitation • Web Application Security (OWASP Top 10) • Network Security (TCP/IP, DNS, Firewalls) • Ethical Hacking • Post-Exploitation • Threat Detection & Analysis • Endpoint Protection • SIEM Concepts • Incident Response • Secure SDLC & DevSecOps • Static Application Security Testing (SAST) • STRIDE Threat Modeling • SSVC / CVSS / EPSS Vulnerability Prioritisation • Adversarial Machine Learning • Identity & Access Management (RBAC) • Cryptography • Defence-in-Depth Architecture • Agile / Scrum

PROFESSIONAL EXPERIENCE

Backend Developer Intern | DIPC Software • Ahmedabad, India (On-site) Jul 2024 – Dec 2024

- Engineered server-side logic and RESTful APIs in a MERN (MongoDB, Express, React, Node.js) stack environment, delivering production-grade backend infrastructure that powered client-facing web applications.
- Implemented secure user authentication and authorization endpoints, enforcing access control policies that protected sensitive user data and prevented unauthorized resource access.
- Designed API integrations between React front-end and MongoDB database with input validation, error handling, and parameterized queries to mitigate injection vulnerabilities and secure data transactions.
- Conducted peer code reviews within an Agile development team using Git for version control, identifying potential security weaknesses and improving code quality, maintainability, and resilience across the SDLC.

MSC PROJECTS

“CWE Insight” — Secure REST API for Vulnerability Intelligence | *ELE8094 Software Assurance*

- Designed and built a Flask REST API that parses MITRE's CWE v4.19.1 dataset (969 weaknesses, ~255K lines of XML) and exposes platform-aware vulnerability analytics across 15 platforms including C/C++, Java, Python, PHP, JavaScript, and .NET.
- Engineered a novel “Consequence-Platform Matrix” endpoint that aggregates CIA-triad impact distribution per platform and ranks highest-blast-radius weaknesses — surfacing CWE-95 Eval Injection as the top-ranked risk for both Java and Python.
- Hardened the XML parser against XXE injection (CWE-611) and Billion Laughs attacks using defusedxml; layered defence-in-depth with allowlist input validation, parameterised SQLite queries, security headers, and rate limiting.
- Designed and committed a 4-stage GitLab CI/CD security pipeline — pytest → Bandit SAST → Safety dependency scan → coverage gate — achieving 90% test coverage against a 70% policy threshold.
- Tech Stack:** Python, Flask, SQLite, defusedxml, Bandit, Safety, pytest, GitLab CI/CD.

AI Security — Adversarial ML Red Team / Blue Team Assessment | *ELE8100 CyberAI*

- Simulated an end-to-end adversarial attack chain against a CNN-based biometric access-control model (FaceGuard): model stealing via proxy-architecture training, untargeted FGSM and PGD evasion sweeps, a targeted impersonation attack (non-authorised user → CEO), and transferability analysis between proxy and victim models.
- Implemented a Blue Team adversarial-training defence and quantified robustness improvement under FGSM and PGD perturbations across multiple ϵ budgets, producing a clean-vs-robust comparison.

- Built the victim CNN in PyTorch (two-block convolutional architecture with batch normalisation and dropout regularisation) on the Olivetti Faces dataset (40 identities, geometric augmentation), achieving ~99% clean test accuracy as the attack baseline.
- **Tech Stack:** PyTorch, NumPy, scikit-learn, Matplotlib; attacks & defences: FGSM, PGD, model stealing, transferability, adversarial training.

Internal Penetration Test — “CyBank” Cryptocurrency Server | *ELE8072 Penetration Testing & Ethical Hacking*

- Conducted a structured black-box internal penetration test against a simulated Linux web server, following NIST SP 800-115 and OWASP Testing Guide methodologies across discovery, enumeration, exploitation, and post-exploitation phases.
- Performed network and service enumeration with Nmap and uncovered hidden development paths using Gobuster; identified an exposed phpinfo() leak and a browsable /development/backups directory containing internal CISO notes.
- Discovered and exploited a critical Remote Command Execution vulnerability (CWE-78 OS Command Injection) in a PHP tool combining a hardcoded password with unsanitised shell_exec(); achieved RCE as the www-data user, enumerated /home, and exfiltrated a sensitive flag from the cryptobank service account.
- Delivered a full client-grade report covering executive summary, evidence-backed findings, risk and business-impact mapping, root-cause analysis, and prioritised remediation; additionally designed an SSVC-based vulnerability prioritisation framework integrating CISA KEV, EPSS, and CVSS signals.
- **Toolkit:** Kali Linux, Nmap (-sn, -sV), Gobuster, manual web reconnaissance, PHP source-code review.

EDUCATION

MSc in Applied Cyber Security | Queen's University Belfast, UK Sep 2025 – Sep 2026
 Modules: Penetration Testing & Ethical Hacking (ELE8072), Software Assurance (ELE8094), CyberAI / AI Security (ELE8100); coverage of network security, cryptography, secure SDLC, STRIDE threat modelling, adversarial ML, and vulnerability prioritisation.

B.Tech, Information Technology Aug 2022 – May 2025
 Silver Oak University, India • Final CGPA: 8.76 / 10

Diploma in Information Technology Aug 2019 – Jul 2022
 Silver Oak University, India • Final CGPA: 7.35 / 10

CERTIFICATIONS

- **CompTIA Security+ (ce)** — CompTIA • Aug 2025 – Aug 2028 • ID: aec4b65f-56bf-4406-b463-ac29b518107f
- **Ethical Hacker** — Cisco Networking Academy • Aug 2025
- **Network Defense** — Cisco Networking Academy • Aug 2025
- **Endpoint Security** — Cisco Networking Academy • Aug 2025
- **CompTIA Security+ (SY0-701) Complete Course & Practice Exam** — Udemy • Aug 2025
- **IT Troubleshooting & Technical Support Helpdesk Bootcamp** — Udemy • Oct 2022

TECHNICAL SKILLS

Cybersecurity: Penetration Testing, Vulnerability Assessment & Exploitation, Ethical Hacking, Web Application Security (OWASP Top 10), CWE / CVE / CVSS analysis, STRIDE Threat Modeling, SSVC Vulnerability Prioritisation, Network Security (TCP/IP, DNS, Firewalls), Endpoint Protection, SIEM, Incident Response, IAM (RBAC), Cryptography, Defence-in-Depth.

Security Tools: Kali Linux, Nmap, Gobuster, Wireshark, Bandit (SAST), Safety (dependency scanning), defusedxml.

AI & Security: Adversarial Machine Learning (FGSM, PGD, Model Stealing, Transferability), Adversarial Training, PyTorch, scikit-learn, Anomaly Detection, Security Analytics.

Programming & Frameworks: Python (Flask, PyTorch, NumPy, defusedxml), JavaScript (Node.js, React), SQL, HTML5, CSS3; Backend: Flask, Express.js, RESTful APIs; Databases: MongoDB, SQLite.

DevSecOps & Methodologies: Git, GitHub, GitLab, GitLab CI/CD, pytest, pytest-cov, Bandit, Safety, Agile / Scrum, Secure SDLC; OS: Linux (Kali, Ubuntu), Windows.